



Nuclear Theory

Winter term 2025/2026

Problem set 1

Deadline to hand in the homework (H (*)) exercise: November 17, 2025, 11:30 via Moodle

November 7, 2025

Problem H1: Lippmann-Schwinger equation (*)

- Using the Schrödinger equation for the scattering of two particles with mass m ,

$$(H_0 + V)|\psi_E\rangle = E|\psi_E\rangle ,$$

where H_0 is the free Hamiltonian, show that the Lippmann-Schwinger equation for the wave function,

$$|\psi_E^\pm\rangle = |\mathbf{k}\rangle + \frac{1}{E - H_0 \pm i\epsilon} V |\psi_E^\pm\rangle ,$$

is satisfied. Here $E = k^2/m$ and the plane wave state satisfies $H_0|\mathbf{k}\rangle = E|\mathbf{k}\rangle$. Why do you need the $\pm i\epsilon$?

- We can define the T -matrix on-shell as the transition matrix that acting on the plane wave state yields the same result as the potential acting on the full scattering state. That is, $T^{(\pm)}(E = k^2/m)|\mathbf{k}\rangle = V|\psi_E^\pm\rangle$. What does it mean that the T -matrix is “on-shell”? (This is a really quick question!)

- Show that matrix elements of the T -matrix satisfy the Lippmann-Schwinger equation

$$\langle \mathbf{k}' | T^{(\pm)}(E) | \mathbf{k} \rangle = \langle \mathbf{k}' | V | \mathbf{k} \rangle + \int d^3p \frac{\langle \mathbf{k}' | V | \mathbf{p} \rangle \langle \mathbf{p} | T^{(\pm)}(E) | \mathbf{k} \rangle}{E - \frac{p^2}{m} \pm i\epsilon} .$$

Are the matrix elements of the T -matrix on the right side on-shell?

- Write the Lippmann-Schwinger equation for the wave function in coordinate space for a local potential $V = V(\mathbf{r})$. To this end, show first that the free Green's function

$$G^\pm(\mathbf{r}', \mathbf{r}; E = k^2/m) = \langle \mathbf{r} | \frac{1}{E - H_0 \pm i\epsilon} | \mathbf{r}' \rangle ,$$

is given by

$$G^\pm(\mathbf{r}', \mathbf{r}; E = k^2/m) = -\frac{m}{4\pi} \frac{e^{\pm ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} .$$

- Show that when the T -matrix is evaluated on-shell, it is proportional to the scattering amplitude, $\langle \mathbf{k}' | T^+(E = k^2/m) | \mathbf{k} \rangle = -\frac{1}{2\pi^2 m} f(k, \theta)$ (with $k' = k$), by analyzing the asymptotic form of the Lippmann-Schwinger equation and comparing to

$$\langle \mathbf{r} | \psi_E^+ \rangle \xrightarrow{r \rightarrow \infty} \frac{1}{(2\pi)^{3/2}} \left(e^{i\mathbf{k} \cdot \mathbf{r}} + f(k, \theta) \frac{e^{ikr}}{r} \right) .$$